

LAPORAN PRAKTIKUM

TEKNOLOGI IOT II

Jaringan Nirkabel dan Komunikasi HTTP



Disusun Oleh:

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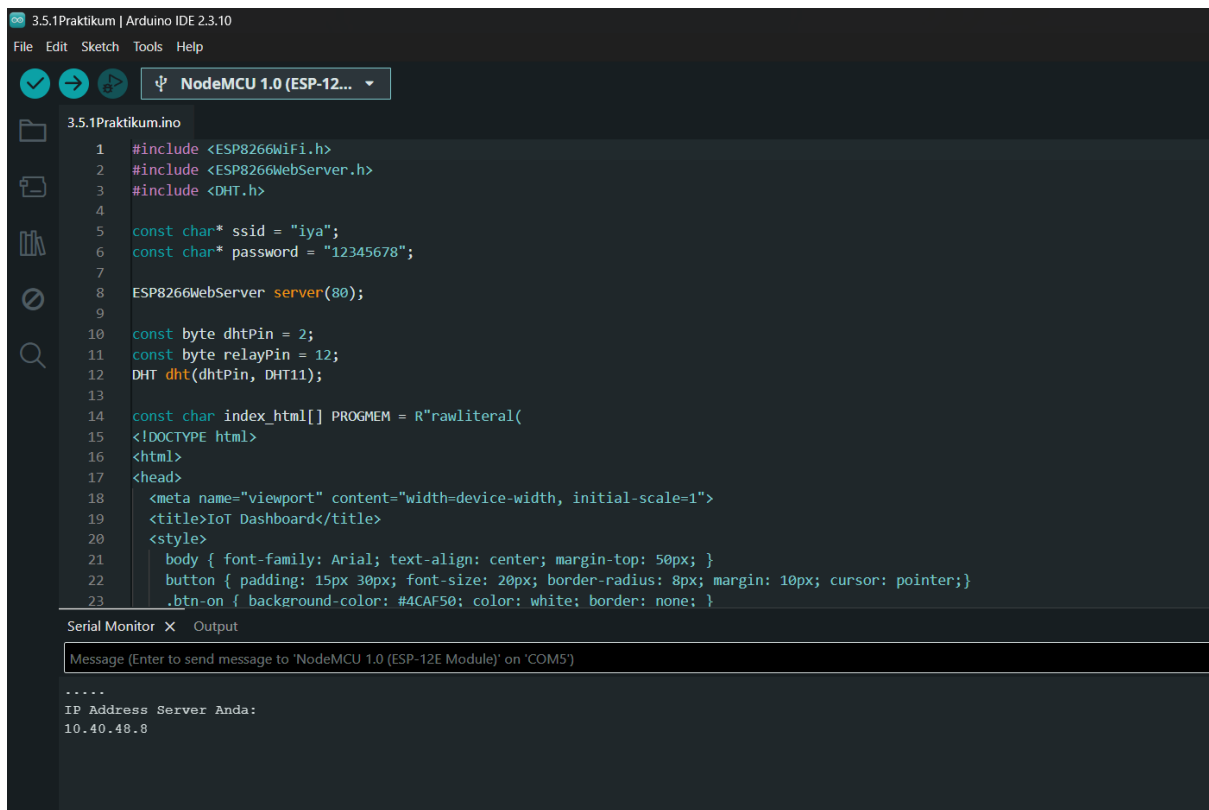
TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

2026/2027

3.5.1 Praktikum 1 (Hafiz)



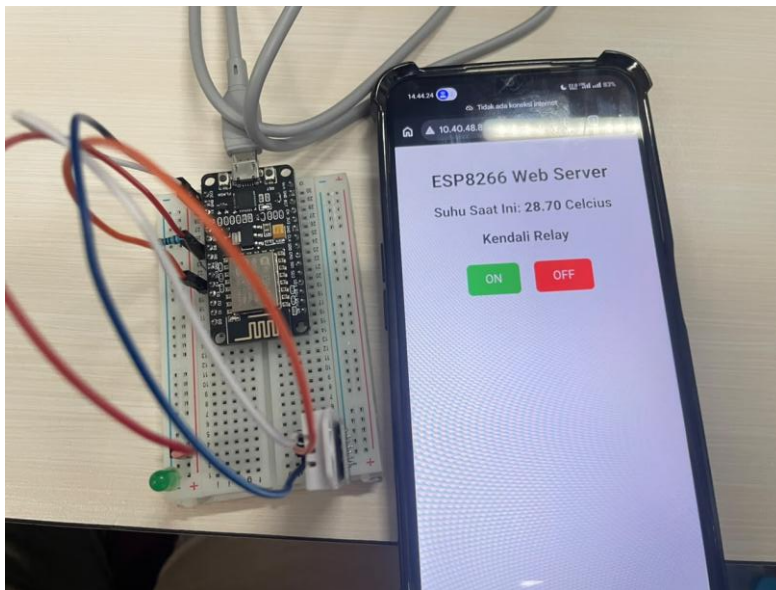
```
3.5.1Praktikum1.ino
1  #include <ESP8266WiFi.h>
2  #include <ESP8266WebServer.h>
3  #include <DHT.h>
4
5  const char* ssid = "iya";
6  const char* password = "12345678";
7
8  ESP8266WebServer server(80);
9
10 const byte dhtPin = 2;
11 const byte relayPin = 12;
12 DHT dht(dhtPin, DHT11);
13
14 const char index_html[] PROGMEM = R"rawliteral(
15 <!DOCTYPE html>
16 <html>
17 <head>
18 <meta name="viewport" content="width=device-width, initial-scale=1">
19 <title>IoT Dashboard</title>
20 <style>
21   body { font-family: Arial; text-align: center; margin-top: 50px; }
22   button { padding: 15px 30px; font-size: 20px; border-radius: 8px; margin: 10px; cursor: pointer; }
23   .btn-on { background-color: #4CAF50; color: white; border: none; }
```

Serial Monitor X Output

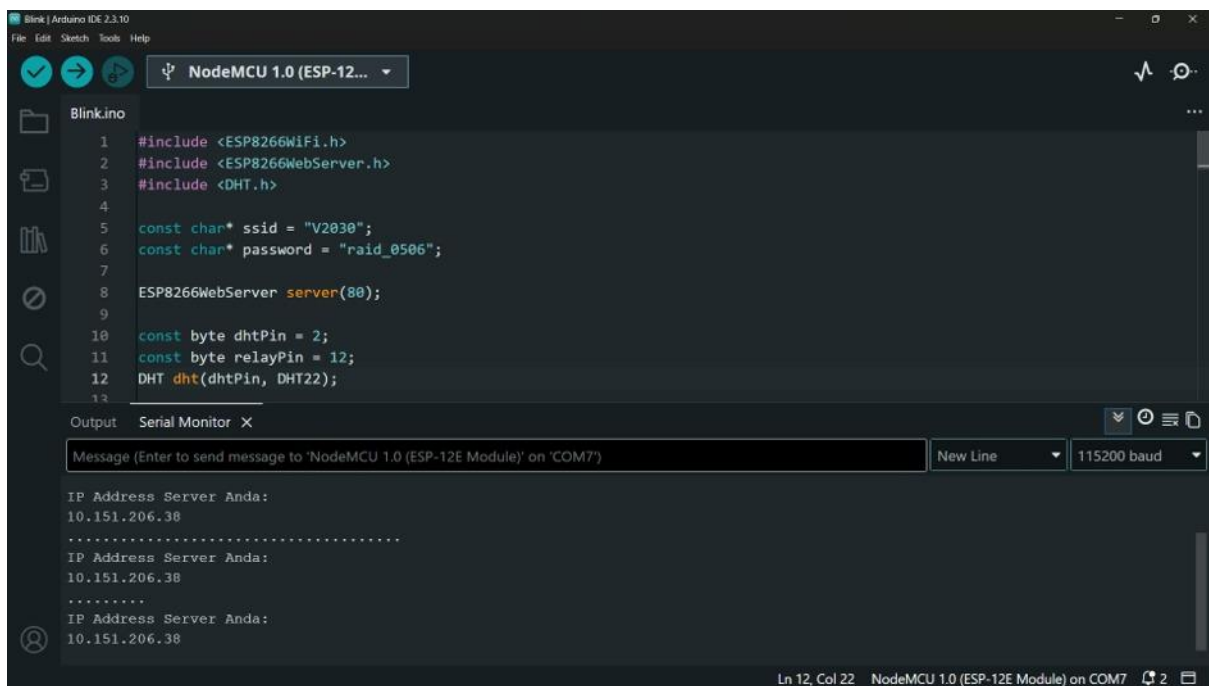
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

.....

IP Address Server Anda:
10.40.48.8



3.5.1 Praktikum 1 (Sadad)



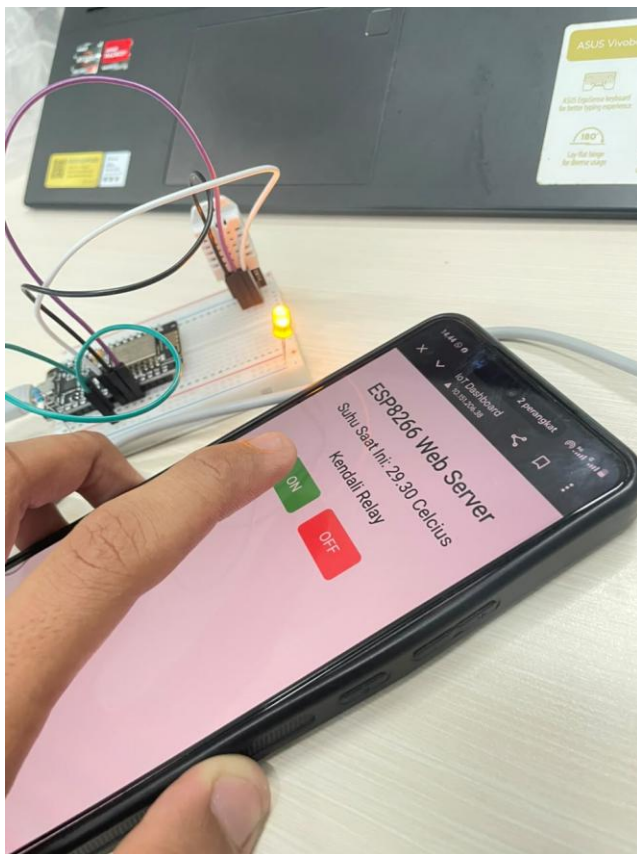
The screenshot shows the Arduino IDE interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. The toolbar has icons for checking, running, and uploading. The target board is set to NodeMCU 1.0 (ESP-12...). The code in the editor is as follows:

```
1 #include <ESP8266WiFi.h>
2 #include <ESP8266WebServer.h>
3 #include <DHT.h>
4
5 const char* ssid = "V2030";
6 const char* password = "raid_0506";
7
8 ESP8266WebServer server(80);
9
10 const byte dhtPin = 2;
11 const byte relayPin = 12;
12 DHT dht(dhtPin, DHT22);
```

The Serial Monitor is open, showing the output of the program. The baud rate is set to 115200. The output shows the IP address of the server being accessed:

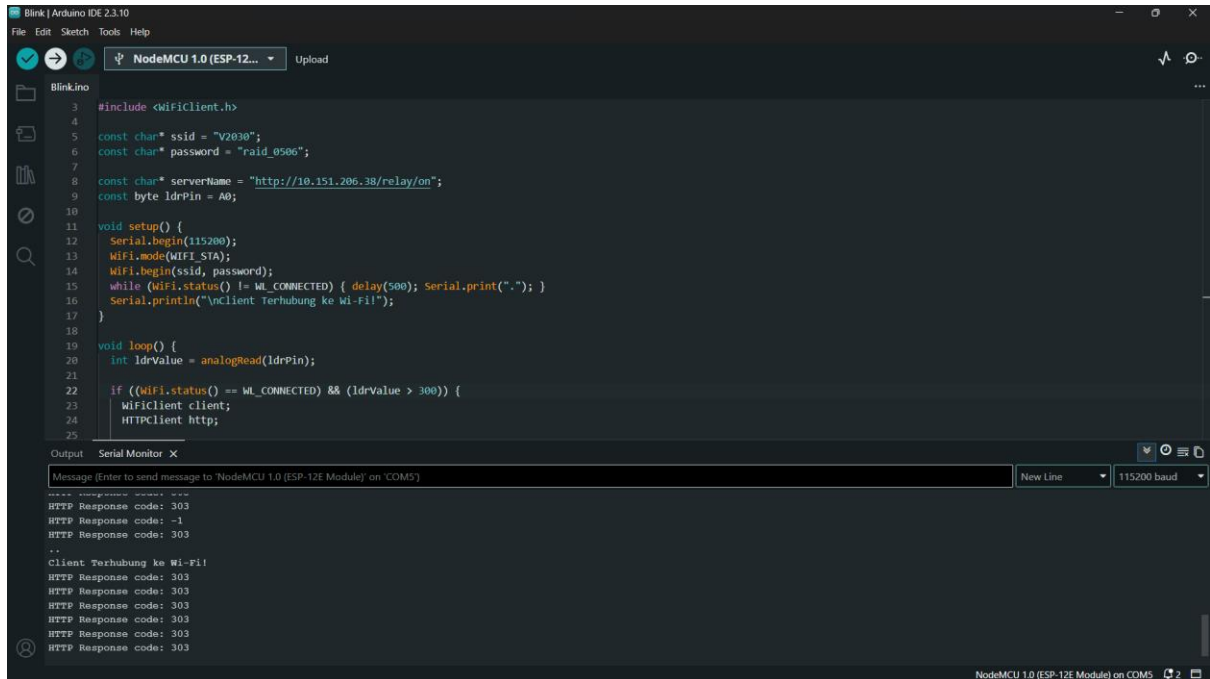
```
IP Address Server Anda:
10.151.206.38
.....
IP Address Server Anda:
10.151.206.38
.....
IP Address Server Anda:
10.151.206.38
```

The status bar at the bottom indicates the current line and column: Ln 12, Col 22. The target board is NodeMCU 1.0 (ESP-12E Module) on COM7.



3.5.2 Praktikum 2

1. client



```
1 #include <WiFiClient.h>
2
3 const char* ssid = "V2030";
4 const char* password = "raid_0506";
5
6 const char* serverName = "http://10.151.206.38/relay/on";
7 const byte ldrPin = A0;
8
9 void setup() {
10   Serial.begin(115200);
11   WiFi.mode(WIFI_STA);
12   WiFi.begin(ssid, password);
13   while (WiFi.status() != WL_CONNECTED) { delay(500); Serial.print("."); }
14   Serial.println("\nClient Terhubung ke Wi-Fi!");
15 }
16
17 void loop() {
18   int ldrValue = analogRead(ldrPin);
19
20   if ((WiFi.status() == WL_CONNECTED) && (ldrValue > 300)) {
21     WiFiClient client;
22     HTTPClient http;
23
24     http.begin(client, serverName);
25     http.GET();
26     Serial.println("HTTP Response code: " + http.getString());
27     http.end();
28   }
29 }
```

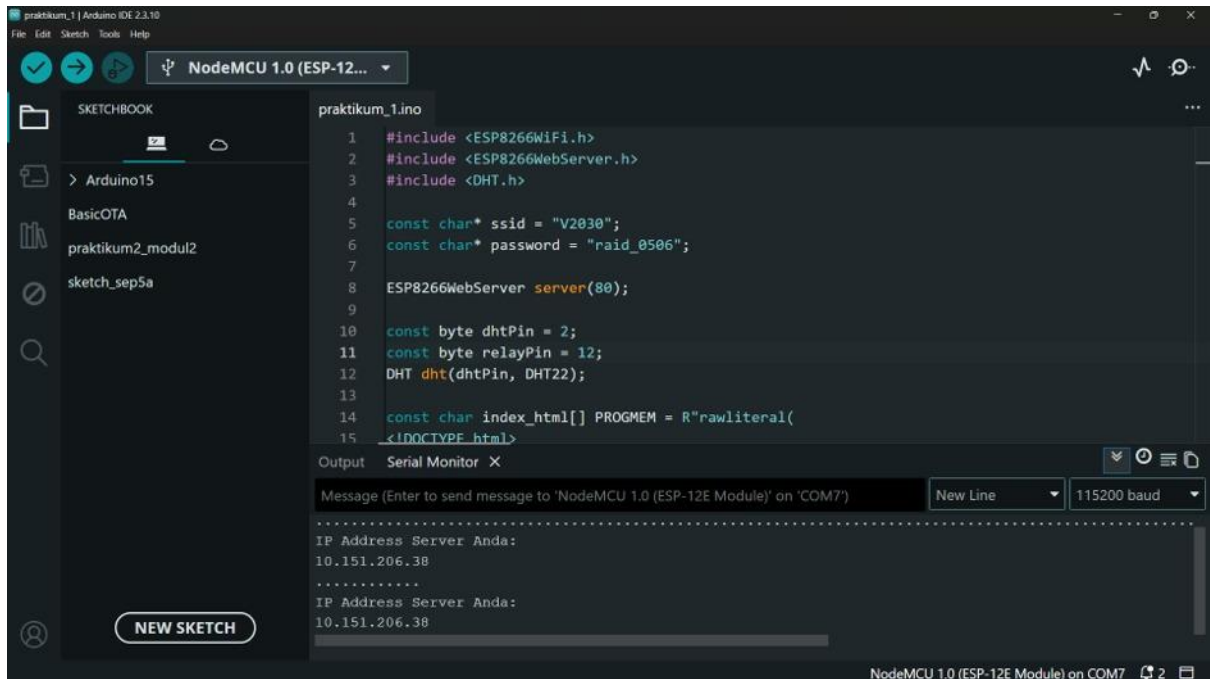
Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

HTTP Response code: 303
HTTP Response code: -1
HTTP Response code: 303
..
Client Terhubung ke Wi-Fi!
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303

NodeMCU 1.0 (ESP-12E Module) on COM5

2. Server



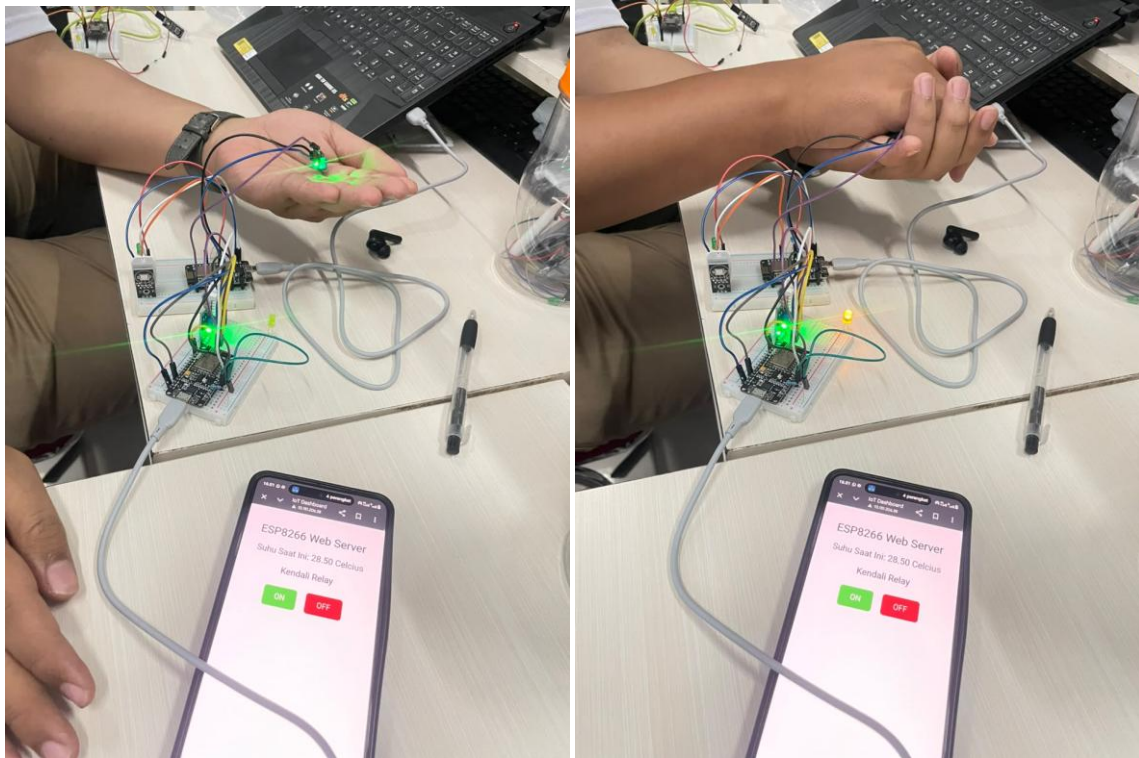
```
1 #include <ESP8266WiFi.h>
2 #include <ESP8266WebServer.h>
3 #include <DHT.h>
4
5 const char* ssid = "V2030";
6 const char* password = "raid_0506";
7
8 ESP8266WebServer server(80);
9
10 const byte dhtPin = 2;
11 const byte relayPin = 12;
12 DHT dht(dhtPin, DHT22);
13
14 const char index_html[] PROGMEM = R"rawliteral(
15 <!DOCTYPE html>
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')

IP Address Server Anda:
10.151.206.38
.....
IP Address Server Anda:
10.151.206.38

NodeMCU 1.0 (ESP-12E Module) on COM7



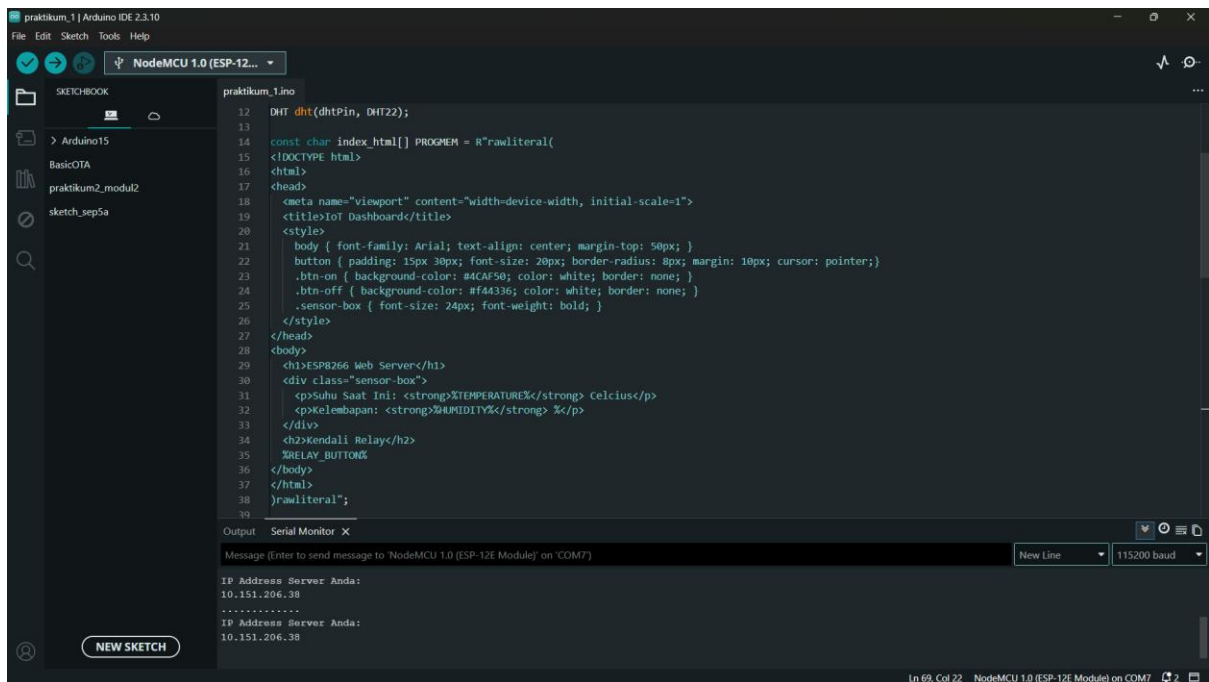
3.6 Latihan

Latihan 1

Mode jaringan Station (STA) digunakan karena pada kedua praktikum NodeMCU berperan sebagai perangkat klien yang terhubung ke jaringan Wi-Fi eksternal, bukan sebagai pemancar jaringan. NodeMCU terhubung ke router atau hotspot HP menggunakan SSID dan password yang sudah tersedia, kemudian memperoleh alamat IP dari jaringan tersebut. Setelah terhubung, browser pada laptop atau HP yang berada di jaringan yang sama dapat mengakses Web Server NodeMCU melalui alamat IP tersebut.

Jika menggunakan mode Access Point (AP), NodeMCU justru harus membuat dan memancarkan jaringan Wi-Fi sendiri agar perangkat lain dapat terhubung langsung kepadanya. Mode tersebut kurang sesuai dengan topologi pada praktikum karena NodeMCU tidak ditugaskan sebagai pembuat jaringan, melainkan sebagai perangkat yang bergabung ke jaringan Wi-Fi yang sudah ada.

Latihan 2



```
praktikum_1 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

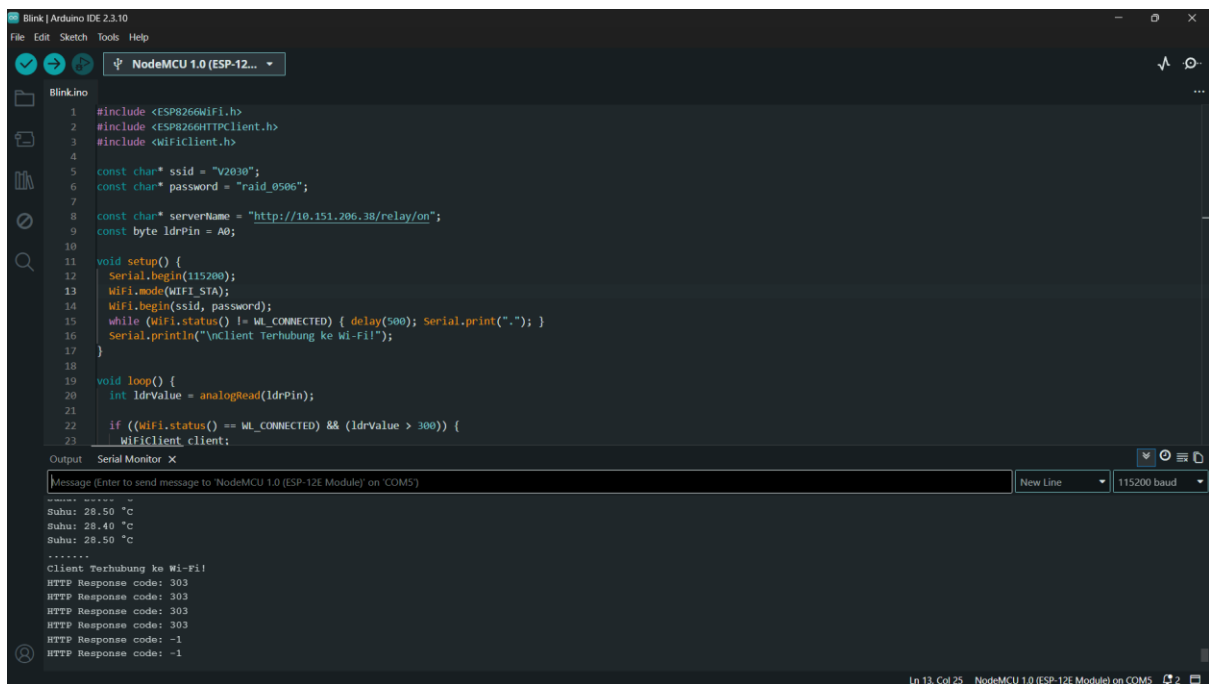
NodeMCU 1.0 (ESP-12...)

praktikum_1.ino
12 DHT dht(dhtPin, DHT22);
13
14 const char index_html[] PROGMEM = R"rawliteral(
15 <!DOCTYPE html>
16 <html>
17 <head>
18 <meta name="viewport" content="width=device-width, initial-scale=1">
19 <title>IoT Dashboard</title>
20 <style>
21 body { font-family: Arial; text-align: center; margin-top: 50px; }
22 button { padding: 15px 30px; font-size: 20px; border-radius: 8px; margin: 10px; cursor: pointer; }
23 .btn-on { background-color: #4CAF50; color: white; border: none; }
24 .btn-off { background-color: #f44336; color: white; border: none; }
25 .sensor-box { font-size: 24px; font-weight: bold; }
26 </style>
27 </head>
28 <body>
29 <h1>ESP8266 Web Server</h1>
30 <div class="sensor-box">
31 <p>Suhu Saat Ini: <strong>%TEMPERATURE%</strong> Celcius</p>
32 <p>Kelembapan: <strong>%HUMIDITY%</strong> %</p>
33 </div>
34 <h2>Kendali Relay</h2>
35 <button type="button" class="btn-on">ON</button>
36 </body>
37 </html>
38 )rawliteral";
39

Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
IP Address Server Anda:
10.151.206.38
.....
IP Address Server Anda:
10.151.206.38

NEW SKETCH
```

Latihan 3



```
Blink | Arduino IDE 2.3.10
File Edit Sketch Tools Help

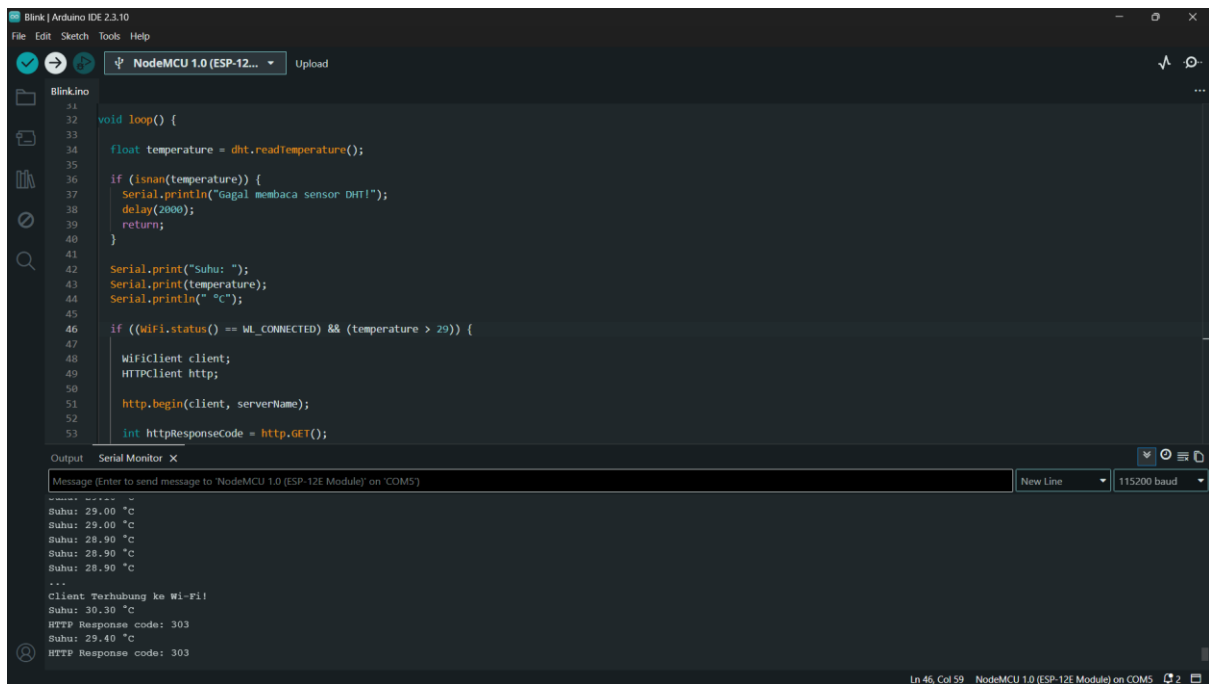
NodeMCU 1.0 (ESP-12...)

Blink.ino
1 #include <ESP8266WiFi.h>
2 #include <ESP8266HTTPClient.h>
3 #include <WiFiClient.h>
4
5 const char* ssid = "V2030";
6 const char* password = "raid_0506";
7
8 const char* serverName = "http://10.151.206.38/relay/on";
9 const byte ldrPin = A0;
10
11 void setup() {
12   Serial.begin(115200);
13   WiFi.mode(WIFI_STA);
14   WiFi.begin(ssid, password);
15   while (WiFi.status() != WL_CONNECTED) { delay(500); Serial.print("."); }
16   Serial.println("\nClient Terhubung ke Wi-Fi!");
17 }
18
19 void loop() {
20   int ldrValue = analogRead(ldrPin);
21   if ((WiFi.status() == WL_CONNECTED) && (ldrValue > 300)) {
22     WiFiClient client;
23     HTTPClient http;
24     http.begin(client, serverName);
25     http.GET();
26     Serial.println("HTTP Response code: " + http.getString());
27     http.end();
28   }
29 }

Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
.....
Suhu: 28.50 °C
Suhu: 28.40 °C
Suhu: 28.50 °C
.....
Client Terhubung ke Wi-Fi!
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: 303
HTTP Response code: -1
HTTP Response code: -1

Ln 13, Col 25 NodeMCU 1.0 (ESP-12E Module) on COM5
```

Latihan 4



```
void loop() {  
  float temperature = dht.readTemperature();  
  
  if (!isnan(temperature)) {  
    Serial.println("Gagal membaca sensor DHT!");  
    delay(2000);  
    return;  
  }  
  
  Serial.print("Suhu: ");  
  Serial.print(temperature);  
  Serial.println(" °C");  
  
  if ((Wifi.status() == WL_CONNECTED) && (temperature > 29)) {  
    WifiClient client;  
    HTTPClient http;  
  
    http.begin(client, serverName);  
  
    int httpResponseCode = http.GET();
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

Suhu: 29.00 °C
Suhu: 29.00 °C
Suhu: 28.90 °C
Suhu: 28.90 °C
Suhu: 28.90 °C
...
Client Terhubung ke Wi-Fi!
Suhu: 30.30 °C
HTTP Response code: 303
Suhu: 29.40 °C
HTTP Response code: 303

Tugas



ESP8266 Web Server

Suhu Saat Ini: **27.70** Celcius

Kelembapan: **47.10** %

Kendali Relay

NYALAKAN (ON)

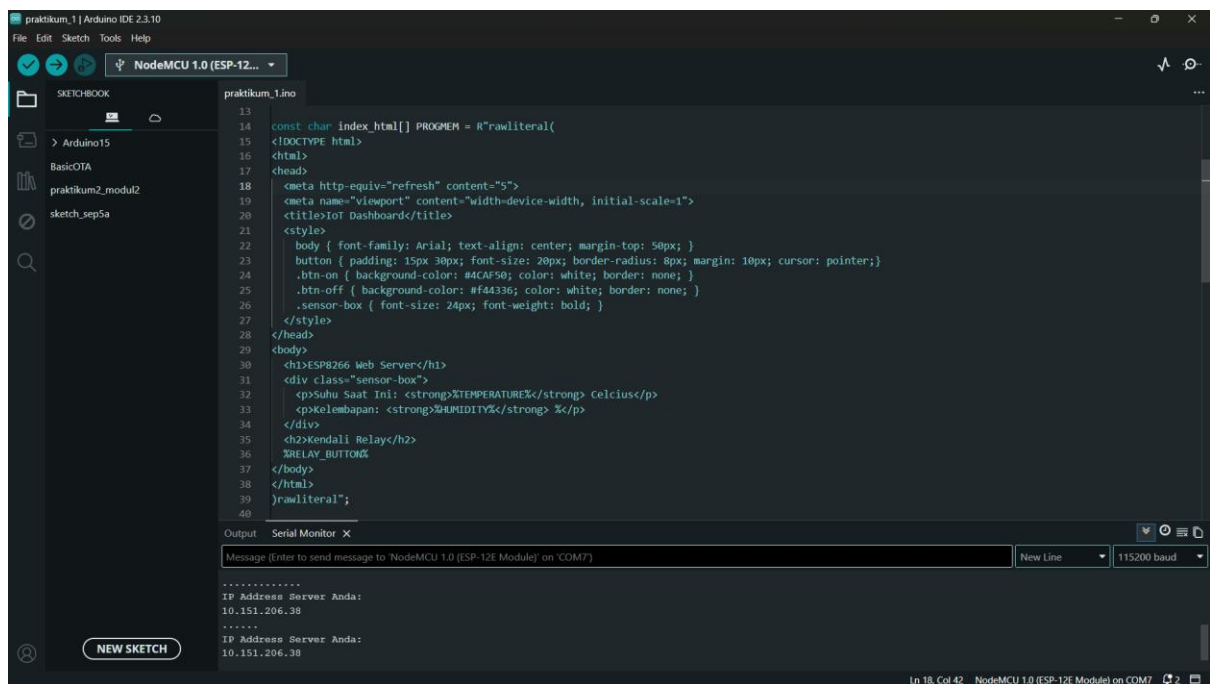
ESP8266 Web Server

Suhu Saat Ini: **27.60** Celcius

Kelembapan: **46.70** %

Kendali Relay

MATIKAN (OFF)



```
13
14 const char index_html[] PROGMEM = R"rawliteral(
15 <!DOCTYPE html>
16 <html>
17 <head>
18   <meta http-equiv="refresh" content="5">
19   <meta name="viewport" content="width=device-width, initial-scale=1">
20   <title>IoT Dashboard</title>
21   <style>
22     body { font-family: Arial; text-align: center; margin-top: 50px; }
23     button { padding: 15px 30px; font-size: 20px; border-radius: 8px; margin: 10px; cursor: pointer; }
24     .btn-on { background-color: #00AEEF; color: white; border: none; }
25     .btn-off { background-color: #E67E22; color: white; border: none; }
26     .sensor-box { font-size: 24px; font-weight: bold; }
27   </style>
28 </head>
29 <body>
30   <h1>ESP8266 Web Server</h1>
31   <div class="sensor-box">
32     <p>Suhu Saat Ini: <strong>TEMPERATURE</strong> Celcius</p>
33     <p>Kelembapan: <strong>HUMIDITY</strong> %</p>
34   </div>
35   <h2>Kendali Relay</h2>
36   <RELAY_BUTTON>
37 </body>
38 </html>
39 )rawliteral";
40
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')

IP Address Server Anda:
10.151.206.38

IP Address Server Anda:
10.151.206.38

Apa kelemahan visual/interaksi (*flickering*) yang Anda alami saat menggunakan metode *Meta Refresh* HTTP konvensional seperti ini untuk sistem yang mengklaim diri sebagai *Real-Time Dashboard*? Tulis evaluasi logis Anda di Laporan Praktikum (Anda bisa membahas masalah *bandwidth*, layar berkedip putih, atau potensi bentrokan jika Anda mengklik tepat di detik ke-5).

Metode Meta Refresh dapat memperbarui data suhu dan kelembapan secara otomatis setiap 5 detik, tetapi memiliki kelemahan jika digunakan sebagai *Real-Time Dashboard*. Setiap kali interval 5 detik tercapai, browser harus memuat ulang seluruh halaman

sehingga dapat menimbulkan efek flickering atau layar berkedip. Hal ini membuat tampilan kurang nyaman dan terasa tidak benar-benar real-time.